

1(a)	<i>idea of</i> replace (live) water plant / <i>Egeria densa</i> with: dead / AW, water plant (of the same volume) or sterile / inert / glass beads / AW, material (of same volume) or pond water with sodium hydrogencarbonate (no water plant) ;	1
1(b)(i)	<i>any seven from:</i> 1 use, lamp / AW, at different distances from plant or use, lamp / AW, with different power ratings / wattages (at same distances) ; 2 use a minimum of five, different / stated, light intensities / AW ; 3 use, a light meter / an app, to measure light intensity ; 4 carry out investigation, in a darkened room / with no other light source / AW ; 5 measure / note / record, the distance the, meniscus / AW, moves along the capillary tube / scale, in a fixed time or measure / note / record, the time the, meniscus / AW, takes to travel, a fixed distance on the capillary tube / scale ; 6 use, same / stated, length of water plant / <i>Egeria densa</i> ; 7 use, same / stated, mass / concentration, of sodium hydrogencarbonate ; 8 use a method to reduce heating effect of lamp ; 9 <i>idea of</i> equilibration / acclimatisation, of, water plant / <i>Egeria densa</i> / whole apparatus (to each light intensity) ; 10 use at least three measurements (at each light intensity / AW,) and calculate a mean ; 11 safety comment with hazard, risk and precaution ;	7

Question	Answer	Marks
1(b)(ii)	<i>horizontal axis labelled:</i> distance from light bulb / light intensity / power rating of light bulb / AW and <i>vertical axis labelled:</i> rate of photosynthesis / rate of movement of meniscus / distance meniscus moved (in set time) / time taken for meniscus to move (a set distance) ; <i>correct units:</i> (distance from light bulb) cm / mm (light intensity) lux / cd (candela) / alternative units (power rating of bulb) W and (rate of photosynthesis) cm min^{-1} / mm min^{-1} / cm s^{-1} / mm s^{-1} (rate of movement of meniscus) cm min^{-1} / mm min^{-1} / cm s^{-1} / mm s^{-1} (distance meniscus moved) cm / mm (time for meniscus to move) min / s ; appropriately shaped line ; positive gradient (may level off at higher light intensities) for, light intensity / power rating of light bulb, against, rate of photosynthesis / rate of movement / distance moved negative gradient (may level off at higher light intensities) for, light intensity / power rating of light bulb against time to move set distance negative gradient (may level off at shorter distances) for distance from light bulb against, rate of photosynthesis / rate of movement / distance moved positive gradient (may level off at shorter distances) for distance from light bulb against time to move set distance	3
1(c)(i)	0.88 ;	1

Question	Answer	Marks
1(c)(ii)	<i>idea of:</i> 1 difficulty in achieving, concentrated / not spread out / AW, spot of extract ; 2 each pigment, spreads out during migration / gives long streak / runs into each other / overlaps / AW (so difficult to measure accurately) ; 3 pigment may, fade / be too pale to see clearly / AW, (so difficult to measure accurately) ; 4 uniformity / quality / grade, of (chromatography) paper ; 5 type of solvent ; 6 different, laboratory / ambient / AW, conditions ;	1

Question	Answer	Marks
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